

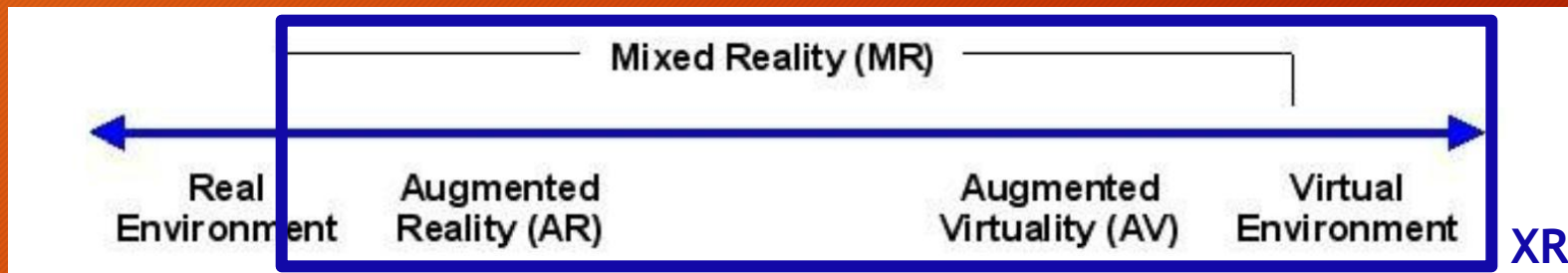
Video Game Programming: XR Game Programming

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X Reality

- XR have been used as acronym for **extended** / **cross** reality and as an umbrella term for:
 - VR, i.e., the simulation of a **virtual** environment
 - AR, i.e., the **augmentation** of real world with virtual elements
 - MR, i.e., a reality which **mixes** real and virtual elements
- More precisely, Milgram et al. proposed the reality-virtuality continuum



- Game programmers can take advantage of XR possibilities to create innovative games

Virtual Reality

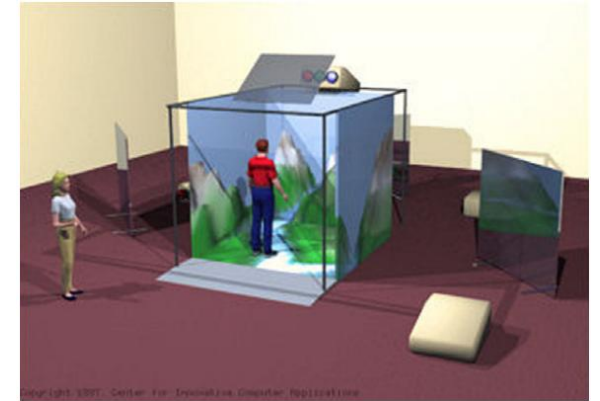
- Every virtual environment (VE) simulation can be considered VR
- VE are often 3D and synthetic, but some use the term also for 360° pictures / videos
- Considering the technology employed, VR can be immersive or non-immersive
- Immersive VR devices are VR headset or multiple projected screens such as CAVEs
- Non-immersive VR devices are 2D screens, such as desktop monitors or smartphone



Oculus Rift SDK and desktop monitor used in a HCI Lab user study

Immersive VR hardware

- Pioneers mostly used expensive CAVEs and headsets (with external 3/6-DOF trackers)
- Consumer headsets with wide FOV and 6-DOF tracking are now available
- Some require a PC, others are standalone
- Smartphone gears enable mobile immersive VR
- Several **VR input devices** are now available:
 - 6-DOF tracked controllers and trackers for different body parts or objects
 - Eye trackers and facial expression trackers
 - Hand tracking cameras and haptic gloves
 - VR treadmills
 - ...



Limniou, Roberts, Papadopoulos, 2008,
Full immersive virtual environment
CAVETM in chemistry education,
Computers & Education, 51(2), 584-593



Quest 3 with tracked controllers

VR sickness

- A well-known problem of immersive VR is **sickness**
- It is mainly due to **visual-vestibular sensory conflict** (see Reason, Brand, Motion Sickness, Academic Press, 1975)
- Sickness in VR games causes players to quit (and negative reviews)
- Low frame rates and continuous virtual locomotion can negatively affect sickness because they increase the sensory conflict
- Optimization and appropriate locomotion techniques are thus fundamental

Quality / optimization balance

- VR should render 3D scene twice (one per eye) at high frame rates
- Some engines support single-pass stereoscopic rendering or can apply automatic optimizations in headsets with eye tracking
- Optimization suggested for mobile mostly applies to VR, but VR players could expect more realism than mobile players
- VR can run from smartphones to workstations, so quality / optimization balance needs specific tuning or adaptive strategies
- Optimized pipelines (e.g., Unity Universal RP) can be useful, but some realistic VR games may need high definition (e.g., HDRP)

Immersive VR locomotion

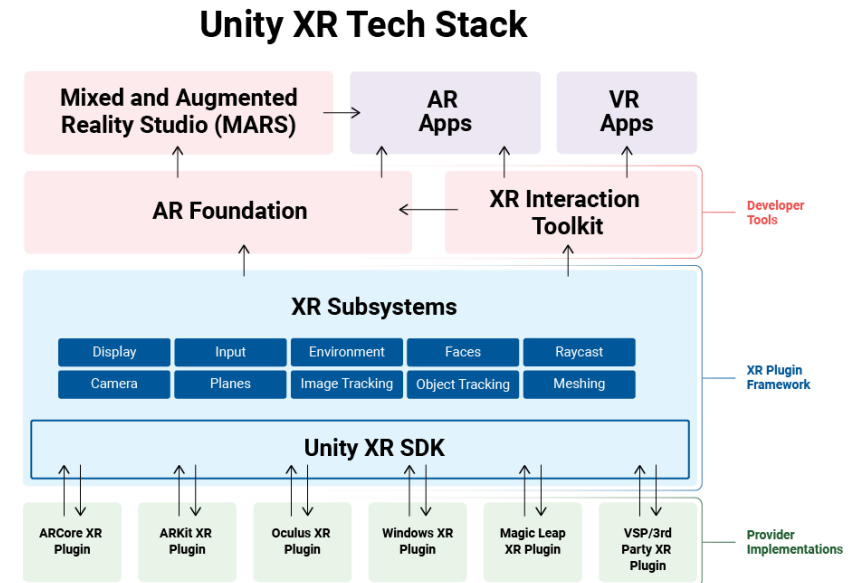
- 19 types of VR locomotion techniques are summarized in Buttussi, Chittaro, 2021, Locomotion in Place in Virtual Reality: A Comparative Evaluation of Joystick, Teleport, and Leaning, IEEE Trans. Vis. CG, 27(1), 125-136
- The paper compares 3 techniques (joystick, teleport, leaning) showing that **teleport** leads to less nausea, better performance, higher usability, and no significant differences in presence
- Instantaneous movement likely reduced sensory conflict
- Several other papers are studying locomotion technique variants
- To reduce conflict using joystick, some games put players in vehicles, keeping some elements at a fixed distance from players

VR GUI

- Diegetic GUIs are preferable in immersive VR games because:
 - non-diegetic GUIs look even less realistic in realistic immersive VEs
 - overlay fixed GUIs can increase sensory conflict in a moving VE
 - in-world diegetic GUIs can allow for natural direct manipulation (e.g., with hands or controllers, touching or pointing)
- If a diegetic GUI cannot be placed in a fixed position in the VE:
 - the GUI can be attached to a controller (or avatar body part)
 - or it can be at a fixed distance (and orientation) from players' head

VR in Unity

- Unity has/had different ways to enable VR game programming:
 - Using third-party plugins (e.g., Oculus and SteamVR ones)
 - Using built-in stereoscopic rendering and tracking build option with old specific packages (up to 2019.4)
 - Using the new XR Tech Stack (in fig.) with 1st- and 3rd-party plugins for specific devices and OpenXR (from 2019.3)



VR input in Unity

- 3rd-party plugins have VR rigs with head and controllers, may offer advanced input features, but are device specific
- Built-in option has limited input features and is now deprecated
- The new XR Tech Stack, includes a unified XR.Input library (in fig.) and the XR Interaction Toolkit package

InputFeatureUsage	Feature type	Legacy input index (left controller/right controller)	Oculus	Vive
primary2DAxis	2D axis	[(1,2)/(4,5)]	Joystick	Trackpad
trigger	Axis	[9/10]	Trigger	Trigger
grip	Axis	[11/12]	Grip	Grip
secondary2DAxis	2D axis	[(17,18)/(19,20)]		
secondary2DAxisClick	Button	[18/19]		
primaryButton	Button	[2/0]	[X/A] - Press	Primary (sandwich button)(1)
primaryTouch	Button	[12/10]	[X/A] - Touch	
secondaryButton	Button	[3/1]	[Y/B] - Press	
secondaryTouch	Button	[13/11]	[Y/B] - Touch	
gripButton	Button	[4/5]	Grip - Press	Grip - Press
triggerButton	Button	[14/15]	Trigger - Press	Trigger - Press
menuButton	Button	[6/7]	Start (left controller only)	
primary2DAxisClick	Button	[8/9]	Thumbstick - Press	Trackpad - Press
primary2DAxisTouch	Button	[16/17]	Thumbstick - Touch	Trackpad - Touch
batteryLevel	Axis			
userPresence	Button		User presence	

Augmented Reality

- Every augmentation of the real world with virtual elements can be considered AR
- Virtual elements are usually (not necessarily) 3D objects
- Earlier proposals were based on QR codes or tags
- Now, more complex detection features are available



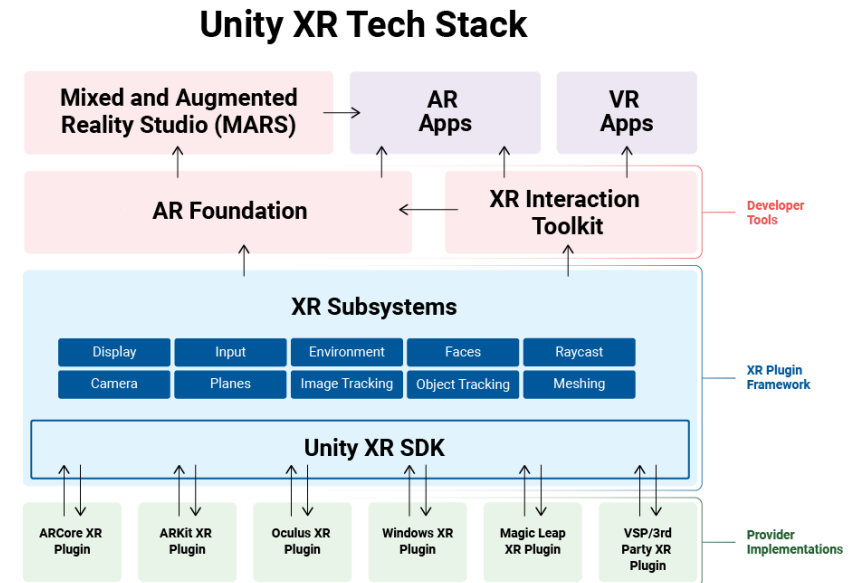
Knightfall™ AR and Angry Birds: FPS

Detection features

- Different AR SDKs support different detection features that can be useful in games:
 - Device tracking tracks device physical position and orientation
 - Plane detection identifies horizontal and vertical surfaces
 - 2D image tracking tracks specific images in the environment
 - 3D object tracking allows to import digital representations of real objects, detect, and track them in the real environment
 - Face and body tracking keeps track of faces and human bodies
 - Gesture recognition raise events when players perform gestures
 - Environment probes detect lighting and color information that are used to apply realistic lights to virtual objects

AR in Unity

- Unity has/had different ways to enable AR game programming:
 - Using third-party SDKs (e.g., ARKit and ARCore ones)
 - Using the new XR Tech Stack (in fig.) with plugins for specific devices and OpenXR (from 2019.3)
- XR Tech Stack includes the AR Foundation package that provides unified access to specific plugins



Useful resources

- Meta for Developer / Horizon (<https://developers.meta.com/horizon>) has both general and Meta-specific instructions
- XR and XR Input sections in Unity manual (<https://docs.unity3d.com/Manual/XR.html>, https://docs.unity3d.com/Manual/xr_input.html) provide detailed information about XR development in Unity
- Unity AR Foundation package manual (<https://docs.unity3d.com/Packages/com.unity.xr.arfoundation@6.5/manual/index.html>) explains how to use AR features
- Unity XR Interaction Toolkit package manual (<https://docs.unity3d.com/Packages/com.unity.xr.interaction.toolkit@3.5/manual/index.html>) describes a toolkit to simplify XR interaction

Bibliography

- Gomila Salas Juan Gabriel, Borromeo Nicolas Alejandro, Hands-On Unity Game Development, 4th ed., Packt Publishing, 2024

